PHYS1408 Exam 2 Study Guide

Chapters 6, 7/8, 9, 10, 11 (plus review of Ch 4–5)

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How the exam is built

The comparison exam (Exam 2, Oct 23, 2015) has 13 multiple-choice problems, each with five answers. Almost every problem is a lightly re-numbered version of an example from the lecture notes. Expect the same here:

Exam 2 #	Problem type	Chapter	Where it is in the notes
1	Halfpipe with a friction patch	7/8	Ch 7/8 p. 4–5
2	Projectile max height from a ledge	4 (review)	—
3	Loop-the-loop minimum speed	6 + 8	Ch 7/8 p. 17–18
4	Sliding up a rough ramp, time to stop	5 (review)	Ch 7/8 p. 7–9 (same physics by energy)
5	Vertical spring launch	7/8	Ch 7/8 p. 17
6	1-D elastic collision	9	Ch 9 p. 2–3
7	Perfectly inelastic, fraction of KE kept (symbolic)	9	Ch 9 p. 1
8	Ballistic pendulum	9	Ch 9 p. 5
9	Hanging mass on a pulley with I	10	Ch 10 p. 4–6
10	Rolling ball, find ω	10	Ch 10 p. 6–8
11	Disk + wheel dropped together (counter-rotating)	11	Ch 11 p. 1, 3
12	Child jumps onto merry-go-round	11	Ch 11 p. 4
13	Block on incline + friction + pulley with I	7/8 + 10	Ch 7/8 p. 13–16

Lecture-note examples that weren't on the 2015 exam but are fair game: flat and banked curves (v_{max} and v_{min}), Atwood's machine, roller-coaster speeds, a spring stopping a sliding mass, power, impulse/force from

a momentum change, the cue ball hitting an 8-ball, parallel axis theorem, rotational kinematics, the rod dropped on a merry-go-round, and a person walking to the center.

Strategy: For each problem, first decide which conservation law applies, then write the before/after equation and solve symbolically before plugging in numbers. If your answer is far from all five choices, recheck the physics, not the arithmetic.

Reading his answer choices. The correct value is often *not* printed exactly. Two rules from the 2015 exam:

1. **Minimum or maximum questions ("how fast must it be moving", "minimum speed", "maximum speed"):** pick the choice that actually *works*, not the closest number. For a minimum, take the smallest choice at or **above** your value. For a maximum, take the largest choice at or **below** it. On Exam 2 #3 (loop, $R = 10$ m), $\sqrt{5gR} = 22.1$ m/s. The answer is d) 23 m/s, because at 21 m/s the mass falls off the loop.
2. **Plain "what is" questions:** take the nearest choice. On Exam 2 #11, the stack turns at 1.86 rad/s, so the answer is e) +2 rad/s (the other neighbor is d) 1.5).

Which tool do I use?

If the problem…	Use
asks for a speed/height and gives start and end positions	**Energy**: $E_i + E_{added} = E_f + E_{lost}$
involves something going in a circle (curve, loop, string)	**Newton's 2nd law** with $\sum F_{toward\ center} = mv^2/r$
has two objects hitting/sticking	**Momentum**: $p_i = p_f$ (plus KE conservation only if *elastic*)
asks for a time or an acceleration	**Forces / kinematics** ($F = ma$, $\tau = I\alpha$)
has something spinning that changes shape or picks up mass	**Angular momentum**: $L_i = L_f$
has something rolling or a pulley with mass	**Energy, with rotational KE** $\frac{1}{2}I\omega^2$ and $v = R\omega$

Symbol key

Every symbol used in this guide is listed here. Each section also defines its symbols right below the equations. "Exam wording" is how the professor's exams and notes refer to the quantity.

Subscripts and modifiers

Mark ($\square$ = any symbol)	Meaning	Example
$\square_i$	**initial**, before the event	v_i = speed at the start
$\square_f$	**final**, after the event	v_f = speed at the end
$\square_0$	starting value (same as $\square_i$)	v_0 = launch speed
$\square_1$, $\square_2$	object 1, object 2	m_1 = mass of object 1
$\square_{1i}$, $\square_{2f}$, …	object number **then** initial/final	v_{2f} = final velocity of object 2
$\square_{max}$, $\square_{min}$	largest / smallest value that still works	v_{max} on a banked curve

Mark (□ = any symbol)	Meaning	Example
Δ ("delta")	**change in** a quantity: final minus initial	$\Delta x = x_f - x_i$
$\vec{\ }$ (arrow on top)	a vector, so direction matters	$\vec{F}$, $\vec{p}$
$\sum$ ("sigma")	**sum of**	$\sum F$ = net force

Constants and material properties

Symbol	Name	Units	Meaning / exam wording
g	gravitational acceleration	m/s^2	always **9.8** m/s^2
μ_s	coefficient of **static** friction	none	grip before sliding starts (curves: tires don't slip)
μ_k	coefficient of **kinetic** friction	none	friction while sliding (ramps, halfpipe patch)
μ	coefficient of friction, type not stated	none	exams often write just "μ"; sliding $\rightarrow$ treat as μ_k, curves $\rightarrow$ μ_s
k	spring constant	N/m	"stiffness" of a Hooke's-law spring

Motion (Ch 2–6)

Symbol	Name	Units	Meaning
m	mass	kg	mass of the moving object (lowercase = the small object)
M	mass	kg	the big object: block of wood, pulley, disk, merry-go-round
v	speed / velocity	m/s	how fast it moves; sign gives direction in 1-D collisions
V	velocity	m/s	velocity of the combined block + bullet in the ballistic pendulum
a	acceleration	m/s^2	rate of change of v
t, Δt	time, time interval	s	Δt in Ch 9 = how long the collision lasts
x, Δx	position, distance moved	m	on an incline, Δx is measured **along the slope**
h	height	m	vertical height above a reference level you choose

Symbol	Name	Units	Meaning
H_{max}	maximum height	m	highest point of a projectile, measured from the ground
d	distance	m	distance moved ($W = Fd$, $E_{lost} = F_f d$)
θ	angle	° or rad	incline/bank angle **from the horizontal**, or launch angle above horizontal
ϕ	angle	°	angle **between** two vectors ($\vec{F}$ and $\vec{d}$, or $\vec{r}$ and $\vec{F}$)
r	radius	m	radius of a circular path; distance of a point mass from the axis
R	radius	m	radius of an object (loop, pulley, disk, ball, merry-go-round)

Forces

Symbol	Name	Units	Meaning
F	force	N ($= \text{kg} \cdot \text{m/s}^2$)	a push or pull
mg	weight	N	force of gravity on mass m
N	normal force	N	surface pushing perpendicular to itself
F_f	friction force	N	$F_f = \mu N$; opposes sliding
T	tension	N	pull of a string
F_c	centripetal force	N	**net** force toward the center, mv^2/r (not a new force)

Energy (Ch 7/8)

Symbol	Name	Units	Meaning
W	work	J	energy transferred by a force: $Fd\cos\phi$
E	total mechanical energy	J ($= \text{kg} \cdot \text{m}^2/\text{s}^2$)	$E = K + U$
K	kinetic energy	J	energy of motion, $\frac{1}{2}mv^2$
K_{rot}	rotational kinetic energy	J	energy of spinning, $\frac{1}{2}I\omega^2$
U	potential energy	J	stored energy
U_g	gravitational PE	J	mgh

Symbol	Name	Units	Meaning
U_s	spring PE	J	$\frac{1}{2}kx^2$
x (springs)	stretch / compression	m	how far the spring is from its relaxed length
E_{added}	energy added	J	work by an outside push ("people pushing the skater")
E_{lost}	energy lost	J	energy removed by friction, $F_f d$
P	power	W (= J/s)	work per unit time
η ("eta")	efficiency	fraction or %	E_f/E_i, the fraction of energy kept (ballistic pendulum)

Momentum (Ch 9)

Symbol	Name	Units	Meaning
p	linear momentum	kg · m/s	$p = mv$; conserved in every collision
p_i, p_f	momentum before / after	kg · m/s	
v_{1i}, v_{2i}	velocities before	m/s	object 1, object 2, **before** the collision (with sign)
v_{1f}, v_{2f}	velocities after	m/s	object 1, object 2, **after** the collision (with sign)
K_i, K_f	KE before / after	J	ratio K_f/K_i = "fraction of KE retained"

Rotation (Ch 10–11)

Symbol	Name	Units	Meaning / exam wording
θ (rotation)	angular position	**rad**	how far it has turned; $\Delta\theta$ = angle turned
s	arc length	m	distance along the rim, $s = R\theta$
ω ("omega")	angular velocity	rad/s	how fast it spins. Exams also call it "**rotational velocity**", "**angular velocity**" or "**rotation rate**"
α ("alpha")	angular acceleration	rad/s^2	how fast ω changes
τ ("tau")	torque	N · m	twisting effect of a force, $\tau = rF\sin\phi$

Symbol	Name	Units	Meaning / exam wording
I	moment of inertia	$\text{kg} \cdot \text{m}^2$	"**resistance to rotation**"; the rotational version of mass
I_{cm} (also I_0)	moment of inertia about the center	$\text{kg} \cdot \text{m}^2$	used in the parallel axis theorem
d (parallel axis)	axis shift	m	distance the axis is moved away from the center
ℓ	length of a rod	m	
L	angular momentum	$\text{kg} \cdot \text{m}^2/\text{s}$	$L = I\omega$; for a point mass moving past at speed v, $L = mvR$

Same letter, two meanings

Read the problem to decide which one is meant:

- L**:** the *length* of a rough patch (halfpipe, Ch 7/8) **or** *angular momentum* (Ch 11).
- θ**:** an *angle of a slope/launch* in degrees **or** the *angle turned* in rotational kinematics, which must be in **radians**.
- d**:** a *distance moved* ($W = Fd$, friction) **or** the *axis shift* in $I = I_{cm} + md^2$.
- x**:** a *position/distance* **or** a *spring's compression*.
- r **vs** R**:** R is the object's own radius; r is where something sits or the radius of a path. A person at the rim has $r = R$.
- m **vs** M**:** in two-object problems, m is the small one (bullet, child, hanging mass) and M the big one (block, merry-go-round, pulley).
- V **vs** v**:** in the ballistic pendulum, v is the bullet and V is block + bullet just after the hit.

Chapter 6 — Circular motion and centripetal force

Key concept

An object moving in a circle at speed v is accelerating toward the center. The *net* force toward the center must equal

$$F_c = \frac{mv^2}{r}.$$

where F_c = centripetal (center-pointing) net force (N), m = mass (kg), v = speed (m/s), r = radius of the circle (m).

mv^2/r is **not** a new force. It is what the real forces (tension, gravity, normal force, friction) have to add up to. The professor draws it as a separate arrow next to the free-body diagram to remind you what $\sum F$ must equal ("no ma!").

Flat curve (friction alone provides F_c)

$$N = mg, \qquad \mu_s mg = \frac{mv^2}{r} \quad \Rightarrow \quad v_{max} = \sqrt{\mu_s g r}.$$

where N = normal force from the road (N), μ_s = coefficient of static friction (no units), $g = 9.8$ m/s^2, r = radius of the curve (m), v_{max} = fastest speed before the car slides outward (m/s).

Mass cancels. Example from the notes: $\mu_s = 0.1$, $r = 50$ m gives $v_{max} = 7$ m/s (15 mph on ice).

Banked curve

Tilt the axes so y is perpendicular to the road and x is along the road. mv^2/r is horizontal, so it has components $\frac{mv^2}{r}\cos\theta$ along the slope and $\frac{mv^2}{r}\sin\theta$ into the slope.

Maximum speed (car wants to slide *up*, so friction points *down* the bank):

$$v_{max} = \sqrt{gr\,\frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}} = \sqrt{gr\,\frac{\sin\theta + \mu_s\cos\theta}{\cos\theta - \mu_s\sin\theta}}.$$

Minimum speed (car wants to slide *down*, so friction points *up* the bank):

$$v_{min} = \sqrt{gr\,\frac{\tan\theta - \mu_s}{1 + \mu_s \tan\theta}} = \sqrt{gr\,\frac{\sin\theta - \mu_s\cos\theta}{\cos\theta + \mu_s\sin\theta}}.$$

where θ = bank angle of the road measured from the horizontal, μ_s = coefficient of static friction, r = radius of the curve (m), v_{max} / v_{min} = fastest / slowest speed at which the car doesn't slide up / down the bank (m/s).

Both forms are equivalent; divide the top and bottom by $\cos\theta$. If $\tan\theta < \mu_s$, then v_{min} doesn't exist: the car can sit still on the bank. Special case $\mu_s = 0$: $v = \sqrt{gr\tan\theta}$ (the only speed that works).

Notes examples: $\theta = 10^\circ$, $\mu_s = 0.1$, $r = 50$ m gives $v_{max} = 11.7$ m/s. With $\theta = 40^\circ$, $\mu = 0.2$, $r = 70$ m, you get $v_{max} = 29.3$ m/s and $v_{min} = 19.4$ m/s.

Vertical loop (combines Ch 6 with energy, Ch 8)

At the top of a loop, both N and mg point toward the center: $N + mg = mv^2/R$. "Barely stays on" means $N = 0$:

$$v_{top} = \sqrt{gR}.$$

Energy from the bottom to the top ($\Delta h = 2R$) gives the **minimum speed at the bottom**:

$$v_{bottom} = \sqrt{5gR}.$$

Starting from rest on a ramp, the **minimum launch height** is

$$h = \tfrac{5}{2}R.$$

Normal force at the bottom (launched from $h = \frac{5}{2}R$): $N - mg = mv^2/R$, which gives $N = 6mg$.

where R = radius of the loop (m), N = normal force from the track (N), v_{top} / v_{bottom} = speed at the top / bottom of the loop (m/s), $\Delta h = 2R$ = height gained from bottom to top, h = starting height of the ramp above the bottom of the loop (m).

Answer-choice trap: $\sqrt{gR}$, $\sqrt{5gR}$ and $\frac{5}{2}R$ are *minimums*. If your value falls between two choices, pick the **higher** one. Anything slower falls off the loop. (On Exam 2 #3, 22.1 m/s gives d) 23, not c) 21.) The same applies to a banked-curve v_{min}. For v_{max}, pick the **lower** choice.

Chapters 7 & 8 — Work, energy, conservation of energy

Definitions

Quantity	Formula	Notes
Work	$W = Fd\cos\phi = \vec{F}\cdot\vec{d}$	F = force, d = distance moved, ϕ = angle between the force and the motion. Only the component of F along the motion does work. A scalar, in joules (J = kg · m²/s²).
Kinetic energy	$K = \frac{1}{2}mv^2$	m = mass, v = speed.
Gravitational PE	$U_g = mgh$	h = height above a reference level you choose; only *changes* matter.
Spring force (Hooke)	$F = -kx$	k = spring constant (N/m), x = stretch or compression from equilibrium (m). The minus sign means the spring pushes back toward equilibrium.
Spring PE	$U_s = \frac{1}{2}kx^2$	Same for stretch or compression.
Power	$P = \dfrac{W}{t} = \dfrac{mgh}{t}$, or $P = Fv$	P = power in watts (J/s), t = time taken (s), v = speed (m/s).

Conservation of energy and the work–energy theorem

$$\boxed{E_i + E_{added} = E_f + E_{lost}} \qquad E = K + U_g + U_s \ (+\ K_{rot})$$

where E_i / E_f = total energy at the start / end (J), E_{added} = energy put in by an outside push, E_{lost} = energy removed by friction, K = kinetic, U_g = gravitational potential, U_s = spring potential, K_{rot} = rotational kinetic energy.

- **No friction, nothing pushing:** $E_i = E_f$.
- **Friction:** $E_{lost} = F_f\, d = \mu_k N d$ (F_f = friction force, d = distance slid on the rough surface, μ_k = coefficient of kinetic friction, N = normal force). On flat ground $N = mg$; on an incline $N = mg\cos\theta$.
- E_{added} is work done by an outside push (a person pushing the skater, a motor).

Standard results (memorize the method, not just the formula)

Situation	Equation	Result
Dropped from rest ("coyote")	$mgh = \frac{1}{2}mv^2$	$v = \sqrt{2gh}$
Roller coaster from rest at h_1	$mg(h_1 - h_2) = \frac{1}{2}mv_2^2$	$v_2 = \sqrt{2g(h_1 - h_2)}$
Halfpipe with a rough patch of length L	$mgh_i = mgh_f + \mu mgL$	$h_f = h_i - \mu L$ (per pass)
Frictionless incline, slide distance Δx up	$\frac{1}{2}v_i^2 = \frac{1}{2}v_f^2 + g\Delta x \sin\theta$	$v_f = \sqrt{v_i^2 - 2g\Delta x \sin\theta}$
Rough incline, sliding **up**		$v_f = \sqrt{v_i^2 - 2g\Delta x(\sin\theta + \mu_k \cos\theta)}$
Rough incline, sliding **down**		$v_f = \sqrt{v_i^2 + 2g\Delta x(\sin\theta - \mu_k \cos\theta)}$
Stopping distance sliding **up** (set $v_f = 0$)		$\Delta x = \dfrac{v_i^2}{2g(\sin\theta + \mu_k \cos\theta)}$
Stopping distance sliding **down** (set $v_f = 0$; only if $\mu_k \cos\theta > \sin\theta$)		$\Delta x = \dfrac{v_i^2}{2g(\mu_k \cos\theta - \sin\theta)}$
Mass slides into a spring, comes to rest	$mgh = \frac{1}{2}kx^2$	$x = \sqrt{2mgh/k}$
Compressed spring launches a mass straight up	$\frac{1}{2}kx^2 = mgh$	$h = \dfrac{kx^2}{2mg}$ (above the release point)
Atwood's machine ($m_1 > m_2$, massless pulley), released from rest	$(m_1 - m_2)gh = \frac{1}{2}(m_1 + m_2)v^2$	$v = \sqrt{\dfrac{2gh(m_1 - m_2)}{m_1 + m_2}}$

Symbols in this table:

- h = height fallen (or the starting height); h_1, h_2 = heights of two points on the track; h_i / h_f = starting / ending height.
- v = speed at the end; v_2 = speed at point 2; v_i / v_f = speed at the start / end.
- L = length of the rough patch (m); μ = its coefficient of friction.
- Δx = distance slid **along the incline** (m); θ = incline angle from the horizontal; μ_k = coefficient of kinetic friction.
- k = spring constant (N/m); x = how far the spring is compressed (m).
- m_1 = heavier hanging mass, which falls; m_2 = lighter hanging mass, which rises.

Sliding-down stopping distance, the professor's way (Ch 7/8 p. 10). Solving $0 = v_i^2 + 2g\Delta x(\sin\theta - \mu_k \cos\theta)$ gives $\Delta x = \dfrac{-v_i^2}{2g(\sin\theta - \mu_k \cos\theta)}$. He then flips the bracket to clear the minus sign: $\Delta x = \dfrac{v_i^2}{2g(\mu_k \cos\theta - \sin\theta)}$. Write it his way so the answer comes out positive. His example ($v_i = 3$ m/s, $\theta = 10^\circ$, $\mu_k = 0.2$) gives $9/[2(9.8)(0.2 \cdot 0.985 - 0.174)] = 19.96$ m. If $\mu_k \cos\theta < \sin\theta$, the block speeds up and never stops.

If the square root comes out negative ("imaginary"), the object stops before covering that distance. Solve for the stopping distance instead. This is a classic exam answer choice: "it stops before it slides the whole _____ m."

Block on an incline tied to a hanging mass (massless pulley)

With m_1 on the incline moving **up** the slope and m_2 hanging and moving **down**:

$$v_f^2 = v_i^2 + \frac{2g\Delta x}{m_1 + m_2}\left(m_2 - m_1 \sin\theta - \mu_k m_1 \cos\theta\right).$$

where m_1 = mass of the block on the incline (kg), m_2 = mass of the hanging block (kg), Δx = distance each block moves (m), θ = incline angle, μ_k = coefficient of kinetic friction between m_1 and the incline, v_i / v_f = shared speed at the start / end.

If m_1 slides **down** the slope while m_2 rises, swap the signs: $(m_1 \sin\theta - m_2 - \mu_k m_1 \cos\theta)$. Friction **always** subtracts, whichever way things move.

Chapter 9 — Linear momentum and collisions

Momentum and force

$$\vec{p} = m\vec{v}, \qquad \vec{F} = \frac{d\vec{p}}{dt} \approx \frac{p_f - p_i}{\Delta t}.$$

where $\vec{p}$ = momentum (kg · m/s), $\vec{F}$ = average force during the hit (N), p_i / p_f = momentum before / after, Δt = how long the contact lasts (s). $\frac{d\vec{p}}{dt}$ means "rate of change of momentum".

Ball bouncing straight back off a wall at the same speed: $p_f = -p_i$, so $|F| = \dfrac{2p}{\Delta t}$. A bounce delivers twice the impulse of a ball that just stops.

Two kinds of collision

	Momentum conserved?	KE conserved?
Inelastic (objects stick = "completely inelastic")	yes	**no**
Elastic	yes	yes

Completely inelastic (they stick):

$$v_f = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}.$$

where m_1, m_2 = the two masses, v_{1i}, v_{2i} = their velocities **before** the collision (with + or − sign for direction), v_f = the shared velocity after they stick.

To find the **fraction of KE kept**, compute K_f/K_i (K_i = total KE before, K_f = total KE after). For symbolic problems (masses like $2m$, speeds like $2v$), the m's and v's cancel. Example: $2m$ at $2v$ hits m at v (same direction), giving $v_f = 5v/3$ and $K_f/K_i = 25/27 \approx 93\%$.

Elastic, 1-D (the "3×5 notecard" formulas):

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2}, \qquad v_{2f} = v_{1i} + v_{1f} - v_{2i}.$$

where v_{1i}, v_{2i} = velocities of objects 1 and 2 **before**; v_{1f}, v_{2f} = velocities **after** (all with signs). Object 1 is whichever one you call object 1; the formulas work either way.

The second formula comes from combining momentum and KE conservation: $v_{1i} + v_{1f} = v_{2i} + v_{2f}$. **Signs matter.** Pick a positive direction and give left-moving objects negative velocities.

Special case, equal masses with one at rest (cue ball hits the 8-ball): the moving ball stops and the struck ball leaves with the full speed.

Ballistic pendulum: two stages, two laws

1. **Collision** (bullet embeds, so momentum is conserved and KE is NOT): $mv = (M + m)V$.
2. **Swing up** (energy conserved): $\frac{1}{2}(M + m)V^2 = (M + m)gh$, so $V = \sqrt{2gh}$.

$$v_{bullet} = \frac{M + m}{m}\sqrt{2gh}, \qquad \frac{K_{after}}{K_{before}} = \frac{(M + m)gh}{\frac{1}{2}mv^2} = \frac{m}{M + m}.$$

where m = bullet mass, M = block mass, $v = v_{bullet}$ = bullet speed before the hit, V = speed of block + bullet just after the hit, h = height the block swings up (m), K_{before} / K_{after} = KE of the bullet before / of block + bullet after (the professor calls this ratio η, the efficiency).

With a realistic bullet ($m \ll M$, "$\ll$" = much less than), the fraction of KE kept is tiny (well under 1%). Never use energy conservation through the collision itself.

Chapter 10 — Rotation

Linear-to-rotational dictionary

Linear	Rotational	Link
x	θ (rad)	$s = R\theta$
v	ω (rad/s)	$v = R\omega$
a	α (rad/s^2)	$a = R\alpha$
m	I	
$F = ma$	$\tau = I\alpha$	$\tau = rF\sin\phi$
$K = \frac{1}{2}mv^2$	$K_{rot} = \frac{1}{2}I\omega^2$	
$p = mv$	$L = I\omega$	
$F = dp/dt$	$\tau = dL/dt$	

Rotational symbols:

- θ = angle turned, in **radians** (rad).
- ω ("omega") = angular velocity in rad/s. Exams also say "rotational velocity" or "rotation rate".
- α ("alpha") = angular acceleration (rad/s^2).
- I = moment of inertia (kg $\cdot$ m^2), "resistance to rotation".
- τ ("tau") = torque (N $\cdot$ m).
- L = angular momentum (kg $\cdot$ m^2/s).
- K_{rot} = rotational kinetic energy (J).
- s = arc length along the rim (m); R = radius of the spinning object (m).
- r = distance from the axis to where the force acts (the lever arm); ϕ = angle between $\vec{r}$ and $\vec{F}$.

Rotational kinematics (constant α), same form as Ch 2

$$\omega_f = \omega_i + \alpha t, \qquad \theta_f = \theta_i + \frac{\omega_i + \omega_f}{2} t, \qquad \omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i), \qquad \theta_f = \theta_i + \omega_i t + \tfrac{1}{2}\alpha t^2.$$

where ω_i / ω_f = angular velocity at the start / end (rad/s), θ_i / θ_f = angle at the start / end (rad), $\theta_f - \theta_i = \Delta\theta$ = angle turned, α = constant angular acceleration (rad/s^2), t = elapsed time (s).

Pick the equation that doesn't need the unknown you aren't given. Notes: $50 \to 100$ rad/s in 10 s gives $\alpha = 5$ rad/s^2. $50 \to 100$ rad/s over 40 rad gives $\alpha = 93.75$ rad/s^2.

Moments of inertia (given on the exam, but know them)

Object	I
Ring / hoop (all mass at R), point mass at distance r	MR^2, mr^2
Solid disk / cylinder	$\frac{1}{2}MR^2$
Solid sphere	$\frac{2}{5}MR^2$
Rod about its middle	$\frac{1}{12}m\ell^2$
Rod about one end	$\frac{1}{3}m\ell^2$

Parallel axis theorem: $I = I_{cm} + md^2$. Example: rod about its end $= \frac{1}{12}m\ell^2 + m(\ell/2)^2 = \frac{1}{3}m\ell^2$. A person (point mass, $I_{cm} \approx 0$) at distance d from the axis has $I = md^2$.

where M or m = mass of the object (kg), R = its radius (m), ℓ = rod length (m), r = distance of a point mass from the axis, I_{cm} = moment of inertia about the object's own center (the professor writes I_0), d = how far the new axis is from that center (m).

In general, $I = \sum m_i r_i^2$: add up each piece's mass m_i times the square of its distance r_i from the axis. Mass farther from the axis means more resistance to rotation.

Hanging mass on a pulley with mass

The torque on the pulley comes from the string tension: $RT = I\alpha$. With $a = R\alpha$:

- Disk pulley ($I = \frac{1}{2}MR^2$): $T = \frac{1}{2}Ma$.
- General pulley: $T = (I/R^2)\,a$.

For the hanging mass, $mg - T = ma$, so

$$a = \frac{mg}{m + I/R^2} \quad \left(= \frac{mg}{m + M/2} \text{ for a disk} \right).$$

where m = hanging mass (kg), M = pulley mass (kg), R = pulley radius (m), I = pulley's moment of inertia, T = string tension (N), a = acceleration of the hanging mass (m/s²), α = angular acceleration of the pulley (rad/s²).

Energy route (often faster; h = distance the mass falls, v_i / v_f = its speed at the start / end): $mgh + \frac{1}{2}(m + I/R^2)v_i^2 = \frac{1}{2}(m + I/R^2)v_f^2$. A pulley with mass acts like extra "effective mass" I/R^2 that has to be sped up but doesn't feel gravity.

Rolling without slipping

$v = R\omega$, so $K_{total} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}\left(m + \frac{I}{R^2}\right)v^2$.

where v = speed of the center (m/s), ω = spin rate (rad/s), R = radius of the rolling object, m = its mass, I = its moment of inertia, K_{total} = total KE (moving + spinning).

Rolling down from height h (vertical drop, m) on an incline of angle θ, starting at rest (a = acceleration along the incline):

Object	v at bottom	a down the incline
Ring	$\sqrt{gh}$	$\frac{1}{2}g\sin\theta$
Disk / cylinder	$\sqrt{\frac{4}{3}gh} = \sqrt{1.33\,gh}$	$\frac{2}{3}g\sin\theta$
Sphere	$\sqrt{\frac{10}{7}gh} = \sqrt{1.43\,gh}$	$\frac{5}{7}g\sin\theta$
Sliding (no friction, no rolling)	$\sqrt{2gh}$	$g\sin\theta$

Then $\omega = v/R$. The radius R matters for ω but not for v. The mass doesn't matter at all. A **sphere beats a disk, which beats a ring**, because less of each object's energy is tied up in spinning.

Chapter 11 — Angular momentum

Conservation of angular momentum

$$L = I\omega, \qquad \boxed{L_i = L_f} \quad \text{when there is no external torque.}$$

where L = angular momentum (kg · m²/s), I = moment of inertia (kg · m²), ω = angular velocity (rad/s), L_i / L_f = total angular momentum before / after. Torque (τ) is a twist from outside the system.

Use it whenever something spinning **picks up mass**, **drops mass**, or **changes shape**. These are "rotational inelastic collisions": KE is generally **not** conserved.

The standard problems

Object dropped onto a spinning disk (they end up turning together):

$$\omega_f = \frac{I_1\omega_{1i} + I_2\omega_{2i}}{I_1 + I_2}.$$

where I_1, ω_{1i} = moment of inertia and starting spin of the bottom disk; I_2, ω_{2i} = the same for the dropped object; ω_f = their shared spin afterward.

If the dropped object is counter-rotating, its ω is negative. If it isn't spinning, $\omega_{2i} = 0$. Notes examples: a disk on a disk gives 1.8 rad/s. A counter-rotating ring gives 1.45 rad/s. A rod from center to edge ($I = \frac{1}{3}m\ell^2$) gives 1.77 rad/s.

Merry-go-round symbols:

- M, R = mass and radius of the merry-go-round (a disk, $I = \frac{1}{2}MR^2$).
- m = the person's mass; r = their distance from the center; v = their running speed (m/s).
- ω_i / ω_f = rotation rate before / after (rad/s).

Person drops onto a merry-go-round at radius r (no running start): $I_{person} = mr^2$, so

$$\omega_f = \frac{\frac{1}{2}MR^2\omega_i}{\frac{1}{2}MR^2 + mr^2}.$$

Person runs and jumps on tangentially at the rim: the person's angular momentum about the axis is $L = mvR$.

$$\omega_f = \frac{\frac{1}{2}MR^2\omega_i + mvR}{\frac{1}{2}MR^2 + mR^2}.$$

If the person runs *against* the rotation, mvR is negative.

Person walks from the rim to the center: I goes from $\frac{1}{2}MR^2 + mR^2$ to $\frac{1}{2}MR^2$, so

$$\omega_f = \frac{M + 2m}{M}\,\omega_i.$$

The merry-go-round speeds up. Its KE *increases*, because the person does work pulling themselves inward.

Energy check: $K = \frac{1}{2}I\omega^2 = \dfrac{L^2}{2I}$. With L fixed, smaller I means more KE.

Review topics that appeared on Exam 2

Projectile from a height (Ch 4)

$$H_{max} = h_0 + \frac{(v_0 \sin\theta)^2}{2g}, \qquad t_{top} = \frac{v_0 \sin\theta}{g}.$$

where H_{max} = highest point above the ground (m), h_0 = height of the ledge it's launched from (m), v_0 = launch speed (m/s), θ = launch angle above the horizontal, $v_0 \sin\theta$ = the vertical part of the launch velocity, t_{top} = time to reach the top (s).

Block sliding on a rough ramp, time to stop (Ch 5)

Going **up**, gravity and friction both oppose the motion:

$$a = g(\sin\theta + \mu_k \cos\theta), \qquad t_{stop} = \frac{v_i}{a}, \qquad d_{stop} = \frac{v_i^2}{2a}.$$

where a = how fast the block slows (m/s²), θ = ramp angle, μ_k = coefficient of kinetic friction, v_i = starting speed (m/s), t_{stop} = time until it stops (s), d_{stop} = distance it slides before stopping (m).

Going **down**: $a = g(\sin\theta - \mu_k \cos\theta)$. If this is negative, the block decelerates.

Common traps

1. **Using energy conservation through a collision.** In a collision, use momentum; save energy for before or after.
2. **Forgetting signs** on velocities or ω's in opposite directions.
3. **Forgetting rotational KE** for rolling objects and massive pulleys.
4. **Treating mv^2/r as an extra force** on a free-body diagram.
5. **Wrong friction direction** on a banked curve: down the bank for v_{max}, up the bank for v_{min}.
6. **Answering with v when asked for ω** (or vice versa). $\omega = v/R$.
7. **Using degrees in rotational kinematics.** θ must be in radians.
8. **Negative under a square root.** It means the object stops first, which is usually an answer choice.
9. **Spring launch height:** measure from the release point (the compressed position), as on Exam 2.
10. **Normal force on an incline** is $mg\cos\theta$, not mg, so friction is $\mu_k mg\cos\theta$.
11. **Rounding a minimum or maximum to the nearest choice.** For a minimum, round *up* to a choice that works. For a maximum, round *down*. (See "Reading his answer choices".)
12. **Choices that aren't spread out.** Neighboring choices can be close (1.5 vs 2 rad/s, 21 vs 23 m/s), so carry three significant figures until the end.

The professor's own wording (for conceptual questions)

On Exam 1, the graded answer was the professor's exact term ("mental model"), not the textbook one ("explanation"). If a conceptual question appears, answer with **his** phrase:

Concept	His wording	Notes
Energy	"Energy is the ability to do work"	Ch 7/8 p. 1
Work	$W = Fd = F_\parallel d = \vec{F}\cdot\vec{d}$; "W is a scalar"; $[W] = [E] = \text{kg}\cdot\text{m}^2/\text{s}^2$ = joule	Ch 7/8 p. 1
Work–energy theorem	$E_i + E_{added} = E_f + E_{lost}$; "lost E is due to friction"; E_{added} "could be people pushing the skater"	Ch 7/8 p. 5
Why use energy	"WET lets us revisit ugly old force and kinematic problems for short solutions"	Ch 7/8 p. 7

Concept	His wording	Notes
Imaginary v_f	"it means block stops before traveling L"	Ch 7/8 p. 8
Hooke's-law spring	"Coils don't touch"; $F = -kx$, $U = \frac{1}{2}kx^2$	Ch 7/8 p. 11
Power	$P = W/t$, J/s = watts; $P = Fd/t = Fv$ ("force at a speed v", the "power to move car")	Ch 7/8 p. 14
Loop	"barely means N = 0"	Ch 7/8 p. 17
Centripetal force	mv^2/r is drawn beside the FBD: "no ma!"	Ch 6 p. 3
Static friction on a curve	$F_f = \mu_s N$ is the "max it can be!"	Ch 6 p. 2
Collisions	"2 types of collision": **inelastic** = p-conservation; **elastic** = p-conservation & K-conservation	Ch 9 p. 1
Elastic formulas	"These go on 3×5 notecard! Good for any 2 objects hitting elastically"	Ch 9 p. 3
Ballistic pendulum	collision = p-cons, swing = E-cons; efficiency $\eta = E_f/E_i$	Ch 9 p. 5
Moment of inertia	"I is resistance to rotation"	Ch 10 p. 2
Rolling	rolling acceleration $\frac{5}{7}g\sin\theta$ is "less than sliding"	Ch 10 p. 3
Angular momentum	"Conservation of Angular Momentum $L_i = L_f$: inelastic collisions"	Ch 11 p. 1
Walking to the center	"It took energy to move from rim to the center" (KE increases)	Ch 11 p. 8

Equation sheet (one page)

Topic	Equations
Circular	$F_c = mv^2/r$; flat curve $v_{max} = \sqrt{\mu_s gr}$; banked $v_{max/min} = \sqrt{gr\frac{\tan\theta \pm \mu_s}{1 \mp \mu_s \tan\theta}}$
Loop	$v_{top} = \sqrt{gR}$; $v_{bottom} = \sqrt{5gR}$; $h_{launch} = \frac{5}{2}R$; $N_{bottom} = 6mg$
Energy	$E_i + E_{added} = E_f + E_{lost}$; $K = \frac{1}{2}mv^2$; $U = mgh$; $U_s = \frac{1}{2}kx^2$; $E_{lost} = \mu_k Nd$
Power	$P = W/t = mgh/t = Fv$
Momentum	$p = mv$; $F = \Delta p/\Delta t$; inelastic $v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$
Elastic	$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2}$; $v_{2f} = v_{1i} + v_{1f} - v_{2i}$
Ballistic	$v = \frac{M+m}{m}\sqrt{2gh}$
Rotation	$v = R\omega$; $a = R\alpha$; $\tau = I\alpha = rF$; $K_{rot} = \frac{1}{2}I\omega^2$; $I = I_{cm} + md^2$
Rot. kinematics	$\omega_f = \omega_i + \alpha t$; $\omega_f^2 = \omega_i^2 + 2\alpha\Delta\theta$; $\Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$

Topic	Equations
Pulley	$a = \frac{mg}{m+I/R^2}$
Rolling	$v = \sqrt{\frac{2gh}{1+I/(mR^2)}}$: ring $\sqrt{gh}$, disk $\sqrt{\frac{4}{3}gh}$, sphere $\sqrt{\frac{10}{7}gh}$
Ang. momentum	$L = I\omega$; $L_i = L_f$; point mass $L = mvr$, $I = mr^2$